

SANGRAMJEET GHOSH

+91 9059150937 | sangramjeet47@gmail.com | [linkedin.com/in/sangramjeetghosh](https://www.linkedin.com/in/sangramjeetghosh) | github.com/JeetInTech | neuroviai.dev
Hyderabad, Telangana, India | B.Tech CSE (AI & ML), Malla Reddy University (2026, GPA: 8.0/10.0)

Technical Skills

Languages	Python (3.10–3.13), TypeScript, SQL
Backend & Cloud	FastAPI, AsyncIO, Pydantic v2, Docker, PostgreSQL, AWS (EC2, S3, Lambda)
Agentic Frameworks	LangGraph (StateGraph), LangChain, Model Context Protocol (MCP SDK), ReAct Agent Loops
LLM APIs & Inference	Google Gemini Live (Bidirectional Audio), Groq (Llama-3), OpenAI API, Ollama (Local RAG), Transformers
Vector DBs & Memory	PostgreSQL (pgvector Cosine Similarity), ChromaDB, SQLite State Machines, Knowledge Graphs
Vision, Media & Audio	OpenCV (FER), GPU Speech Transcription, FFmpeg CLI, Librosa, PyAudio, Kokoro TTS, Playwright

Professional Experience

Freelance Software Engineer Aug 2025 – Present
Independent Remote

- Architected multimodal autonomous agents utilizing LangGraph state machines, Gemini Live streaming, and custom tool suites.
- Built Model Context Protocol (MCP) servers and persistent RAG pipelines with sub-10ms semantic retrieval across vector stores.

Data Science & Backend Intern Mar 2025 – Dec 2025
Inikola Technologies Hyderabad, India

- Deployed FastAPI backend endpoints serving natural language scoring and feature extraction models under strict latency budgets.
- Built automated data validation pipelines using Pydantic and PostgreSQL to sanitize and index unstructured text feeds.

Featured Projects

Sebastian — Autonomous AI Operating System & Private Digital Butler github.com/JeetInTech
Python, Gemini Live API, Kokoro Neural TTS, PostgreSQL, pgvector, Model Context Protocol, PyAudio

- Built a full-duplex, 25-module voice operating system via Gemini Live (~400ms turnaround) with barge-in interruption.
- Orchestrated 92 registered tools across Green/Yellow permission gates via `npx` child-process MCP servers (GitHub, Playwright, Filesystem).
- Engineered turn-by-turn semantic memory retrieval powered by **pgvector cosine similarity (sub-10ms)** and persistent Knowledge Graph.

Genesis: Autonomous Multi-Board Job Application Agent github.com/JeetInTech
Python, Playwright, Groq LLM API, SQLite, BeautifulSoup4, Jinja2, pdflatex Engine

- Developed an autonomous recruitment agent executing end-to-end applications across 5 job portals via Playwright and CDP.
- Parsed job listings with Groq Llama-3, dynamically generated tailored LaTeX resumes, and compiled PDFs via local `pdflatex` in 1.8s.
- Managed autonomous multi-step questionnaire answering and form filling with SQLite finite state tracking and screenshot audit trails.

Virtual Therapist — LangGraph Autonomous Multi-Modal Agent github.com/JeetInTech
Python, LangGraph, ChromaDB, OpenCV (FER), Librosa, Groq LLM, Pydantic v2

- Implemented a LangGraph StateGraph ReAct loop orchestrating **24 deterministic domain tools** and episodic conversational memory.
- Combined real-time facial micro-expression recognition (FER/OpenCV) and acoustic voice prosody analysis (Librosa) into an affective state vector.
- Built deterministic safety guardrails intercepting crisis indicators in <50ms via rule-based checks that run before any LLM call.

Neuroviai Multi-Model LLM Routing & Gateway Engine neuroviai.dev
Python, FastAPI, Pydantic v2, Groq, Google Gemini Pro, HuggingFace, Local Ollama

- Designed a 4-tier circuit-breaking LLM gateway (Gemini Pro → Groq → HuggingFace → Ollama) achieving 99.7% uptime under 12k tokens/min rate limits.
- Implemented exponential backoff with jitter on HTTP 429 and Pydantic v2 schema validation on model outputs.

Genesis: Video Pipeline — Autonomous YouTube Shorts Agent github.com/JeetInTech
Python, Ollama, Groq LLM, Edge-TTS, FFmpeg CLI, YouTube Data API v3, OAuth 2.0

- Built an autonomous agent on a daily schedule that generates scripts, synthesizes voiceovers, and composites video via FFmpeg with zero manual intervention.
- Sustained continuous unattended operation for 100+ days, autonomously publishing 138 Shorts via the YouTube Data API v3.

Education

Bachelor of Technology in Computer Science and Engineering (AI & ML) Graduated May 2026
Malla Reddy University, Hyderabad, Telangana **GPA: 8.0 / 10.0**